

Worked Example (1)

At a tensile stress of 100 MPa, 50% of cylindrical specimens made of an engineering ceramic fibre will fail. Assuming a Weibull factor of $m=5$, calculate:

- The stress at which only 10% of the ceramic specimens will fail.
- The stress at which only 10% of the ceramic cylindrical specimens will fail, if the diameter of the specimen is half of the initial diameter.

$$P_s(V_0) = \exp \left\{ - \left(\frac{\sigma}{\sigma_0} \right)^m \right\} \qquad P_s(V) = \exp \left\{ - \frac{V}{V_0} \left(\frac{\sigma}{\sigma_0} \right)^m \right\},$$

$$\ln P_s(V_0) = - \left(\frac{\sigma}{\sigma_0} \right)^m \qquad \ln P_s(V) = - \frac{V}{V_0} \left(\frac{\sigma}{\sigma_0} \right)^m.$$

$$\text{i. } P(V_0) = e^{-\left(\frac{P_0}{6_0}\right)^n}$$

$$\therefore -\left(\frac{100}{6_0}\right)^5 = \ln 50\% \quad \therefore 6_0 = 107.61 \text{ (MPa)}$$

$$\therefore -\left(\frac{6'}{6_0}\right)^5 = \ln 90\% \quad \therefore 6' = 68.61 \text{ (MPa)}$$

$$\text{ii. } P(V) = [P(V_0)]^n = e^{-\frac{V}{V_0} \left(\frac{P_0}{6_0}\right)^n}$$

$$\therefore \frac{V}{V_0} = \frac{h \cdot \pi \left(\frac{1}{2} \cdot \frac{1}{2} d^2\right)}{h \cdot \pi \left(\frac{1}{2} d^2\right)} = \frac{1}{4}$$

$$\therefore -\frac{1}{4} \left(\frac{P_0}{6_0}\right)^5 = \ln 90\%$$

$$\therefore 6 = 90.53 \text{ MPa}$$

Worked Example (1)

- i. The stress at which only 10% of the ceramic specimens will fail.

$$P_s(V_0) = \exp \left\{ - \left(\frac{\sigma}{\sigma_0} \right)^m \right\} \quad \ln P_s(V_0) = - \left(\frac{\sigma}{\sigma_0} \right)^m .$$

We find σ_0 based on the data we got from the testing samples. Rearranging the equation below:

$$\ln(1/P_s) = (\sigma/\sigma_0)^m \rightarrow \sigma_0 = \sigma / [\ln(1/P_s)]^{1/m} =$$

$$\sigma_0 = 100 \text{ MPa} / [\ln(2)]^{1/5} = 100 \text{ MPa} / [\ln(2)]^{0.2} = 107.6 \text{ MPa}$$

With $\sigma_0 = 107.6 \text{ MPa}$, $P_s = 0.9$ and $m = 5$.

$$\sigma = \sigma_0 * [\ln(1/P_s)]^{1/m} = 107.6 * [\ln(1/P_s)]^{1/5} = 107.6 * [\ln(1/0.9)]^{0.2} = 68.6 \text{ MPa}$$

Worked Example (1)

ii. The stress at which only 10% of the cylindrical ceramic specimen will fail, if the diameter of the specimen is half of the initial diameter.

$$P_s(V) = \exp \left\{ -\frac{V}{V_0} \left(\frac{\sigma}{\sigma_0} \right)^m \right\}, \quad \ln P_s(V) = -\frac{V}{V_0} \left(\frac{\sigma}{\sigma_0} \right)^m.$$

$$\ln(P_s(V)) = -V/V_0(\sigma/\sigma_0)^m$$

$$\ln(P_s(V_1)) = -V_1/V_0(\sigma_1/\sigma_0)^m$$

$$\ln(P_s(V_2)) = -V_2/V_0(\sigma_2/\sigma_0)^m$$

$$\text{But } \ln(P_s(V_1)) = \ln(P_s(V_2)) = 0.9; \text{ hence: } V_1/V_0(\sigma_1/\sigma_0)^m = V_2/V_0(\sigma_2/\sigma_0)^m$$

$$\text{Simplifying: } V_1\sigma_1^m = V_2\sigma_2^m$$

$$\text{If } V_{\text{cylinder}} = 1/4 L \pi d^2; d_1^2 \sigma_1^m = d_2^2 \sigma_2^m;$$

$$\sigma_2 = [(d_1/d_2)^2]^{1/m} \cdot \sigma_1; \sigma_2 = [(2)^2]^{1/m} \cdot \sigma_1 = [4]^{0.2} \cdot 68.6 = \mathbf{90.5 \text{ MPa}}.$$

Mechanical Properties of Ceramics

Statistics of strength and Weibull distribution

- Note this assumes that component is subjected to a uniform tensile stress
- In many applications, it isn't constant, but varies with position, so:

$$P_s(V) = \exp \left\{ -\frac{1}{\sigma_0^m V_0} \int_V \sigma^m dV \right\}.$$

Worked Example (2)

Stalactite hanging downwards from the roof of a cave.

The geometry is idealised as a cone of length L and semiangle α . The cone angle is assumed small so that the base radius equals αL . The stalactite density is ρ .

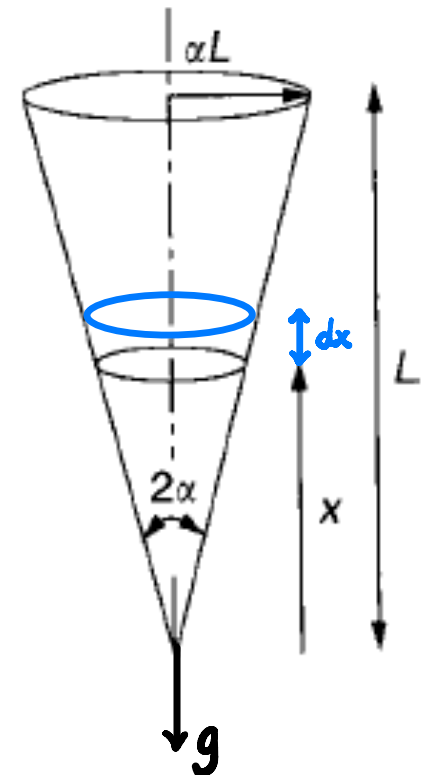
- (a) *Show that the variation of tensile stress σ experienced by the stalactite as a function of the height x is given by*

$$\sigma = \frac{\rho g x}{3}$$

N.B. Volume of a cone is given by $(1/3) \times \pi \times (\text{base radius})^2 \times \text{height}$.

- (b) *By using expression for Weibull statistics with a varying stress, Show that the probability of survival $P_s(L)$ for a stalactite of length L is given by:*

$$P_s(L) = \exp \left\{ - \left(\frac{\rho g}{3 \sigma_0} \right)^m \frac{\pi \alpha^2 L^{m+3}}{(m+3) V_0} \right\}$$



$$(a) \quad b = \frac{F}{A} = \frac{mg}{A} = \frac{\rho V g}{A} = \frac{\rho \left(\frac{1}{3} x (\pi a x^2) \right) g}{\pi a x^2} = \frac{\rho g x}{3}$$

$$(b) \quad P_s(L) = e^{-\frac{\rho}{3} \left(\frac{b}{a} \right)^m} = e^{-\frac{1}{3} \frac{\rho}{a^m} \int_V b^m dV}$$

$$\text{For } \int_V b^m dV$$

$$\int_V b^m dV = \int_V \left(\frac{\rho g x}{3} \right)^m dV = \left(\frac{\rho g}{3} \right)^m \int x^m [dx \pi (a x^2)]$$

$$= \left(\frac{\rho g}{3} \right)^m \frac{\pi a^2}{3} \int x^{m+2} dx$$

$$\text{At } L, \quad x = L \quad \therefore P_s(L) = e^{-\left(\frac{\rho g}{3 a^m} \right)^m \frac{\pi a^2 L^{m+3}}{(m+3) V_0}}$$

Worked Example (2)

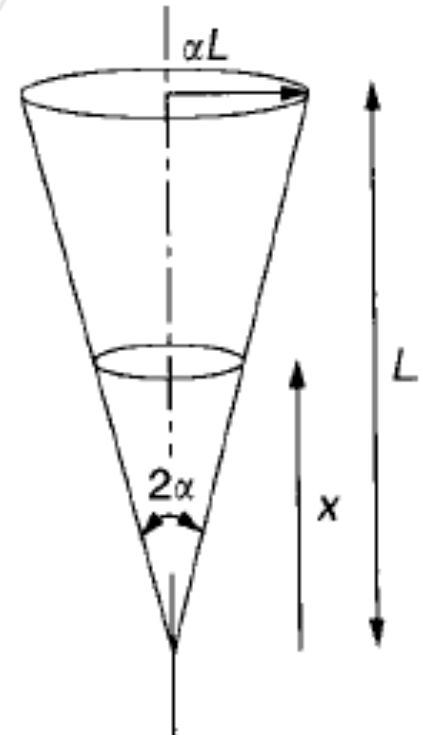
N.B. Volume of a cone is given by $(1/3) \times \pi \times (\text{base radius})^2 \times \text{height}$.

(a) Weight of material below section at x is

$$\text{Gravitational Force: } \rho g \times \frac{\pi}{3} (\alpha x)^2 x.$$

$$\text{Cross-sectional area} = \pi (\alpha x)^2.$$

$$\sigma = \frac{\rho g x}{3}$$



Worked Example (2)

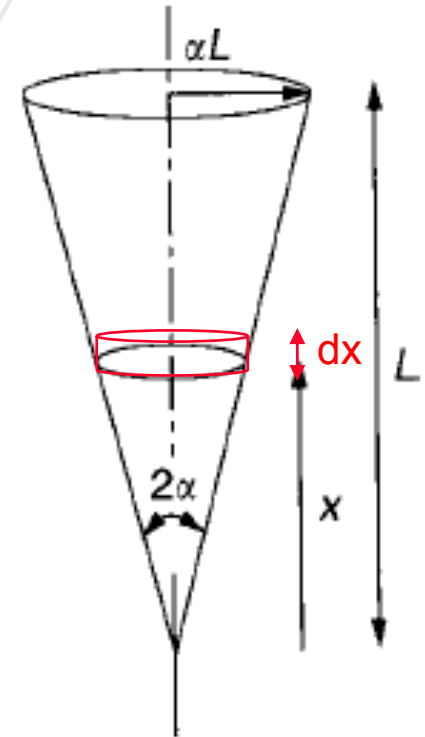
Consider a disc of thickness dx , with volume dV

$$dV = \pi(\alpha x)^2 dx$$

The earlier equation, and substituting

$$P_s(L) = \exp\left(\int_v -\left(\frac{\sigma}{\sigma_0}\right)^m \frac{dV}{V_0}\right)$$

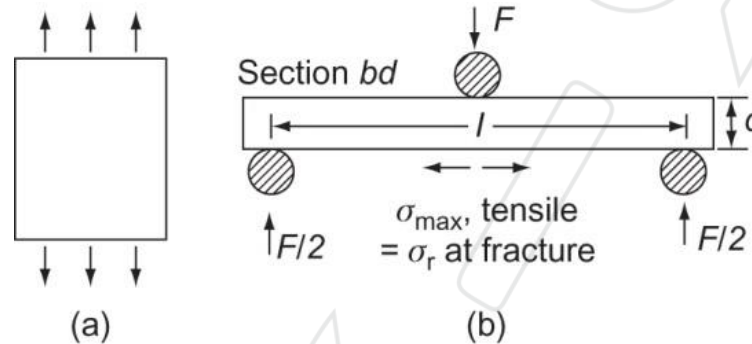
$$P_s(L) = \exp\left(\int_0^L -\left(\frac{\rho g x}{3\sigma_0}\right)^m \frac{\pi(\alpha x)^2 dx}{V_0}\right)$$



Integrating gives

$$P_s(L) = \exp\left\{-\left(\frac{\rho g}{3\sigma_0}\right)^m \frac{\pi\alpha^2 L^{m+3}}{(m+3)V_0}\right\}.$$

Worked Example (3)



Modulus of rupture tests are carried out on a glass microscope slides (loaded and supported in their faces, not their edges). The width of the slides, w , is much greater than the thickness. The distance between the support rollers is l .

Note that the survival probability, $P_s(A)$, of a ceramic associated with surface defects, when under constant stress σ , is given by (see Fig (a) above):

$$P_s(A) = \exp \left\{ -\frac{A}{A_0} \left(\frac{\sigma}{\sigma_0} \right)^m \right\}$$

Whilst that for a surface defect ($P_s(A)$) under varying stress, σ , is given by (see Fig (b) above): $P_s(A) = \exp \left\{ -\frac{1}{\sigma_0^m A_0} \int_A \sigma^m dA \right\}$

Show that the stress integral for the microscope slide in the expression directly above is given by the equation on the side, where m is the Weibull modulus for the glass and σ_r is the modulus of rupture:

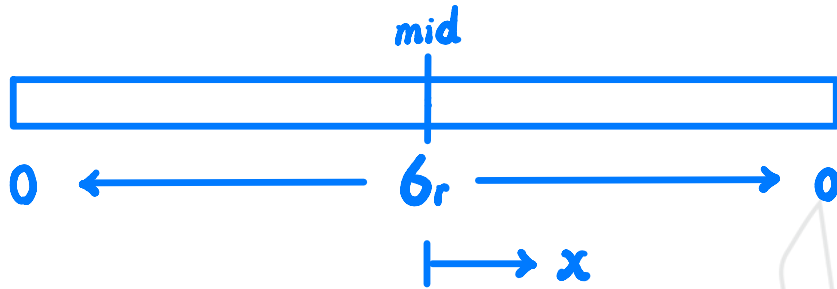
$$\int_A \sigma^m dA = \frac{wl\sigma_r^m}{(m+1)}$$

Hence, show that the tensile strength of the glass is related to the modulus of rupture by the following equation:

$$\sigma_{TS} = \frac{\sigma_r}{\{2(m+1)\}^{1/m}}$$

Hence calculate the ratio σ_r/σ_{TS} for $m=10$.

$$\int_A b^m dA = \int_{\frac{1}{2}l}^l (b_r \frac{\frac{1}{2}l - x}{\frac{1}{2}l})^m dA \quad (x \geq 0) + \int_{-\frac{1}{2}l}^0 (b_r \frac{\frac{1}{2}l + x}{\frac{1}{2}l})^m dA \quad (x \leq 0)$$



$$\therefore \int b^m dA = 2 \int_0^{\frac{1}{2}l} (b_r \frac{\frac{1}{2}l - x}{\frac{1}{2}l})^m (w dx) = 2w b_r^m \int_0^{\frac{1}{2}l} (1 - \frac{2x}{l})^m dx$$

$$= 2w b_r^m \left[\frac{1}{m+1} (-\frac{1}{2}) (1 - \frac{2x}{l})^{m+1} \right]_0^{\frac{1}{2}l} = \frac{w l b_r^m}{m+1}$$

$$P_s(A) = e^{-\frac{1}{b_0^m \lambda_0} \int b^m dA} = e^{-\frac{A}{\lambda_0} (\frac{b_{75}}{b_0})^m}$$

$$\therefore - \frac{1}{6_o^m A_o} \int 6^m dA = - \frac{A}{A_o} \left(\frac{6_{Ts}}{6_o} \right)^m$$

$$\therefore \int 6^m dA = A 6_{Ts}^m$$

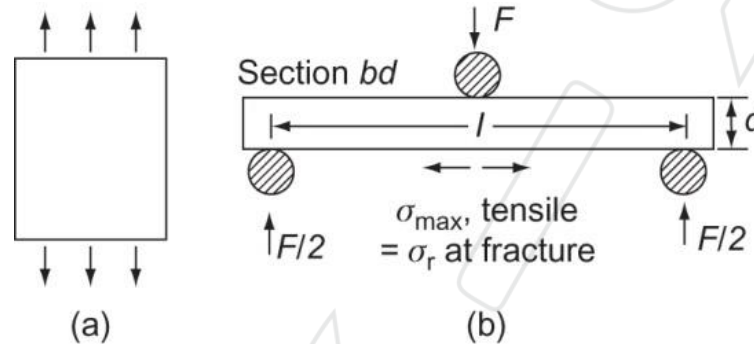
$$\therefore \frac{w l 6_r^m}{m+1} = A 6_{Ts}^m = 2 w l 6_{Ts}^m$$

$$\therefore 6_{Ts} = \frac{6_r}{(2m+2)^{\frac{1}{m}}}$$

When $m = 10$,

$$\frac{6_r}{6_{Ts}} = (2 \cdot 10 + 2)^{\frac{1}{10}} = 1.3622$$

Worked Example (3)



The maximum tensile stress occurs at mid span, where it is defined as σ_r . If we go away from midspan towards one end of the slide, by distance x , the surface stress decreases linearly, and is given by:

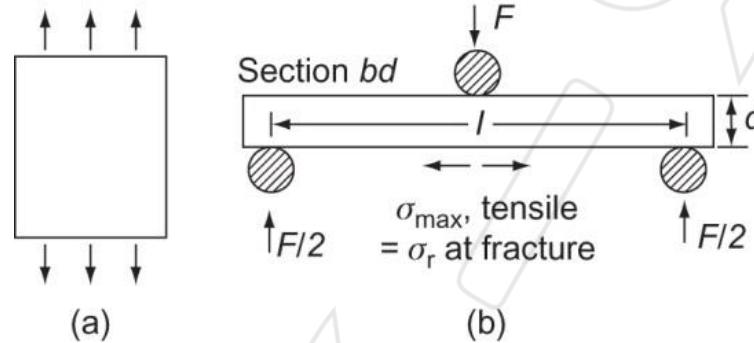
$$\sigma(x) = \sigma_r \left(\frac{\ell/2 - x}{\ell/2} \right) = \sigma_r \left(1 - \frac{2}{\ell} x \right)$$

In order to form the stress integral, we consider a surface element $dA = wdx$, i.e. a narrow strip going transversely across the width of the bottom face where σ can be considered to be constant. Then,

$$\int_A \sigma^m dA = 2 \int_0^{\ell/2} \sigma_r^m \left(1 - \frac{2}{\ell} x \right)^m w dx = 2 \sigma_r^m w \int_0^{\ell/2} \left(1 - \frac{2}{\ell} x \right)^m dx$$

Note that $\int (a + bx)^m dx = \frac{(a + bx)^{m+1}}{(m+1)b}$

Worked Example (3)



$$\int_0^{l/2} \left(1 - \frac{2}{l}x\right)^m dx = \left[\left(1 - \frac{2}{l}x\right)^{m+1} \frac{1}{(m+1)} \left(-\frac{l}{2}\right) \right]_0^{l/2} = -\frac{1}{2(m+1)} \left[l \left(1 - \frac{2}{l}x\right)^{m+1} \right]_0^{l/2}$$

$$= -\frac{1}{2(m+1)} \left\{ l \left(1 - \frac{2}{l} \frac{l}{2}\right)^{m+1} - l \left(1 - \frac{2}{l} 0\right)^{m+1} \right\} = -\frac{1}{2(m+1)} (-l) = \frac{l}{2(m+1)}$$

$$\int_A \sigma^m dA = \frac{2\sigma_r^m w l}{2(m+1)} = \frac{w l \sigma_r^m}{(m+1)}$$

Worked Example (3)

In a tensile test on the slides, the stress is a constant at $\sigma = \sigma_{TS}$. The area under tension is $A = 2wl$. Therefore, the given equation becomes:

$$P_S(A) = \exp \left\{ - \frac{2wl}{A_0} \left(\frac{\sigma_{TS}}{\sigma_0} \right)^m \right\}$$

In a bending test, the earlier equation becomes:

$$P_S(A) = \exp \left\{ - \frac{1}{\sigma_0^m A_0} \frac{wl \sigma_r^m}{(m+1)} \right\}$$

Equating these two survival probabilities for both tests gives:

$$\frac{2wl}{A_0} \left(\frac{\sigma_{TS}}{\sigma_0} \right)^m = \frac{1}{\sigma_0^m A_0} \frac{wl \sigma_r^m}{(m+1)}$$

$$\sigma_{TS}^m = \frac{\sigma_r^m}{2(m+1)}$$

$$\sigma_{TS} = \frac{\sigma_r}{\{2(m+1)\}^{1/m}}$$

$$\frac{\sigma_r}{\sigma_{TS}} = \{2(m+1)\}^{1/m} = \{2(10+1)\}^{1/10} = 1.36$$